

From Connectivity to Care: How Digital Inclusion Shapes Women's Healthcare Decision-Making in Bangladesh

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Abstract

Digital connectivity has rapidly expanded in South Asia, but its implications for women's autonomy in healthcare remain underexplored. This study introduces a novel digital inclusion index linking Internet, Finance, and Technology (BD-LIFT). This study develops a 0–3 cumulative index (BD-LIFT) capturing digital inclusion through smartphone ownership, internet use, and mobile money adoption. The study utilizes nationally representative data from the 2022 Bangladesh Demographic and Health Survey and employs multi-level logistic regression to examine how digital inclusion influences women's ability to participate in decisions regarding their own healthcare. The results of the current study show that, compared to women without full BD-LIFT access, those with full access were significantly more likely to report healthcare autonomy (OR = 2.40, $p < 0.001$), even after adjusting for wealth, education, employment, and residence. Cluster-level variation (ICC ≈ 0.08) indicates that communities influence autonomy, yet digital inclusion remains a consistent predictor across different contexts. The findings from the current study are significant to research, practice, and policy. The study demonstrates that telecommunications infrastructure and mobile financial platforms are not only economic drivers but also enablers of gender equity and health empowerment, linked to the Sustainable Development Goals (SDG 3 and SDG 5). The research provides a replicable, data-driven BD-LIFT that advances big data analytics, quality of service, and human-computer interaction research, offering the communications field a scalable tool to evaluate how digital systems improve social outcomes.

CCS Concepts

• Human-centered computing;

Keywords

Digital inclusion, women's autonomy, mobile health, telecommunications, and Sustainable Development Goals

ACM Reference Format:

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1 INTRODUCTION

Mobile phones and internet services have expanded quickly in South Asia, creating new opportunities for social inclusion. Yet women in low- and middle-income countries remain 19% less likely than men to use mobile Internet, and two-thirds of the 900 million women still offline live in South Asia and Sub-Saharan Africa [1], [2], [3], [4], [5]. In Bangladesh, this divide is particularly pronounced, with rural women facing compounded disadvantages in smartphone ownership, internet use, and mobile financial services [2]. These inequities reach beyond connectivity and limit women's autonomy in education, employment, and healthcare [2], [6]. Healthcare decision-making autonomy is widely recognized as a cornerstone of women's empowerment. Studies show that women who participate in healthcare decision-making are more likely to use preventive services, adhere to treatment, and achieve improved maternal and child health outcomes [3]. However, in Bangladesh, fewer than half of ever-married women report having a say in their own healthcare, with even lower rates among rural and low-income groups. This underscores that the digital divide is not only a technological barrier but also a structural determinant of health equity [6], [7], [8]. The current study addresses a key question: to what extent does digital inclusion predict women's autonomy in healthcare decision-making in Bangladesh? This study utilizes a nationally representative dataset from the Bangladesh Demographic and Health Survey 2022 (N = 15112). It constructs a novel composite measure, BD-LIFT (Bangladesh Digital Life Inclusion for Females & Technology), that captures the cumulative impacts of smartphone ownership, internet use, and mobile financial activity. Furthermore, the study advances multilevel modeling to quantify both individual- and community-level effects. The contribution is threefold: (i) introducing a replicable, technology-grounded digital inclusion index, (ii) demonstrating how telecommunications infrastructure and mobile finance directly enhance women's healthcare agency, and (iii) providing policy-relevant evidence linking digital inclusion to Sustainable Development Goals (SDG3: health; SDG5: gender equality).

2 RELATED WORKS

Digital inclusion has become a central theme in communications and development research, reflecting how access to mobile devices, internet services, and financial platforms reshapes social and health

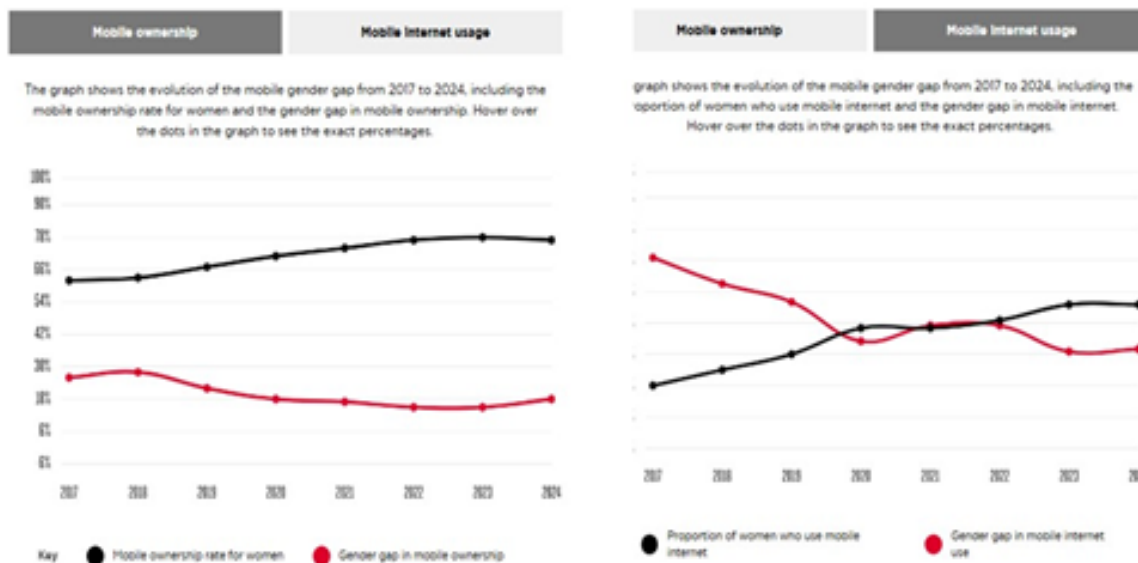


Figure 1: Women’s mobile ownership and mobile internet use, 2017–2024, showing persistent gender gaps. Source: GSMA Mobile Gender Gap Report 2024 [1]

outcomes. Global indices, such as the ICT Development Index and the GSMA Mobile Gender Gap Report, indicate gains in connectivity; however, gaps persist in terms of gender, geography, and income [1]. Women in low- and middle-income countries remain 19% less likely than men to use mobile Internet, and nearly two-thirds of the 900 million women who remain offline live in South Asia and Sub-Saharan Africa, where smartphone ownership and financial inclusion gaps remain wide [1]. Figure 1 illustrates these trends by showing that while women’s mobile ownership has risen steadily since 2017, the gender gap in ownership has narrowed only slightly. In contrast, women’s mobile internet use lags, with the gender gap in usage remaining stubbornly high and even widening in some years. These patterns show that aggregate measures of connectivity hide persistent disparities, especially in Bangladesh. They also emphasize the need for context-specific tools, such as the BD-LIFT index, which can capture progressive digital inclusion, including smartphone ownership, internet use, and mobile money, and directly link to women’s healthcare autonomy and broader gender equity outcomes.

Several empirical and mixed-method research studies have highlighted the role of digital divides in shaping women’s reproductive health and decision-making power in the South Asia region [8], [9], [10]. Several empirical research studies show that women’s autonomy in healthcare decision-making remains limited, with cultural norms, wealth disparities, and education levels acting as significant determinants [5], [6], [11], [12], [13], [14], [15], [16], [17]. Complementing this, Mostafiz et al. use the 2017–18 Bangladesh Demographic and Health Survey (BDHS) to examine autonomy outcomes, finding that education, partner characteristics, and household resources strongly condition women’s ability to make health-related choices [18]. Yet these studies rely primarily on socioeconomic variables, failing to directly capture digital inclusion, despite the

rapid spread of smartphones and mobile financial technologies in Bangladesh during the same period. Thus, while autonomy has been extensively studied in public health, the explicit integration of telecommunications and digital finance into this discourse remains underdeveloped. Bangladesh stands at a crossroads in its digital transformation, with technological advances, expanding access to remote and rural areas, and enhanced mobile connectivity and internet access [5], [16], [19], [20], [21], [22]. While the past decade has witnessed the rapid expansion of mobile connectivity, recent evidence suggests a reversal. According to the Bangladesh Telecommunication Regulatory Commission (BTRC), mobile subscribers decreased by 6 million, and mobile internet users fell by 13.2 million between June 2024 and January 2025 [2], [23], [24]. This decline reflects rising living costs, higher taxes, and political instability, which have reduced households’ ability to sustain mobile use. By January 2025, the total number of mobile internet users had decreased to 116 million, down from 129.2 million just seven months earlier. Scholars and policymakers describe this contraction as alarming, warning that it threatens the momentum of digitalization and exacerbates existing inequalities, especially for women in rural and low-income households.

Taken together, the literature reveals three consistent themes that frame the contribution of the current study. First, prior indices of digital access remain primarily global or macro-level, often focusing on infrastructure coverage or network expansion without validating micro-level outcomes of autonomy. Second, although South Asian and Bangladeshi studies have richly explored women’s healthcare autonomy, they have not incorporated telecommunications or mobile finance variables into their frameworks, leaving an empirical gap for testing digital inclusion as a determinant. Third, methodological work in computer science demonstrates the potential of machine learning and big data analytics to capture digital

divides; however, few studies have adapted these approaches to nationally representative survey data with policy-relevant health outcomes. The novel BD-LIFT index from the current study directly addresses these gaps by creating a cumulative, micro-validated measure of digital inclusion based on smartphone ownership, internet use, and mobile financial activity. It is applied to the 2022 BDHS, which surveyed over 30,000 ever-married women aged 15–49, providing a robust, nationally representative sample [4]. The current study demonstrates both the individual-level significance of digital access and community-level variation, as captured by cluster random effects, by linking this index to a multilevel model of women's healthcare autonomy. In doing so, it advances the literature across both communications and health, providing a replicable framework for integrating telecommunications research with gender equity and development policy.

3 METHODOLOGY

This study draws on the Bangladesh Demographic and Health Survey (BDHS) 2022, a nationally representative household survey conducted by the National Institute of Population Research and Training (NIPORT) and fieldwork by Mitra and Associates from June to December 2022. The survey employed a two-stage stratified cluster sampling design, covering ever-married women aged 15–49 years across all eight administrative divisions of Bangladesh. The analytic sample for this study consists of 30,078 respondents, drawn from the Individual Recode (IR) file. Survey weights were applied to adjust for sampling design and ensure national representativeness. The dependent variable is autonomy in healthcare decision-making. Following DHS coding [4], women were asked: "Who usually makes decisions about the respondent's healthcare?". Responses were dichotomized into autonomous (1) if the woman decided alone or jointly with her spouse, and not autonomous (0) if the husband or other family members solely made decisions. This binary outcome has been widely used in the autonomy literature, including prior multilevel studies in Bangladesh and Ethiopia [18], [25], [26]. The key independent variable is the Bangladesh Digital Life Inclusion for Females & Technology (BD-LIFT) index, a four-level cumulative measure of digital inclusion (0 = no device; 1 = device only; 2 = device + Internet; 3 = device + Internet + mobile money). Its stepwise structure captures progressive engagement with digital ecosystems and their effects on healthcare autonomy. Covariates include education (five attainment levels), household wealth quintile (poorest to richest, based on the DHS asset index), age (15–19, 20–29, 30–39, 40–49), employment status (working vs. not working), and residence (urban vs. rural). These factors are consistently linked to autonomy in South Asia; partner, religion, and fertility variables were excluded to reduce collinearity and preserve parsimony. Variables were recoded for interpretability, and missing data (<5%) were handled via complete-case analysis. Survey weights were applied to adjust for the DHS's complex survey data collection design.

In the current study, a multilevel logistic regression model is used to account for both individual- and community-level variation in women's healthcare autonomy. The outcome is variable (Y_{ij}) coded 1 if a woman reported having a say in her own healthcare and 0 otherwise. The primary predictor was the BD-LIFT index of

digital inclusion, modeled both independently and in interaction with residence (urban versus rural). Additional controls included household wealth quintiles, respondents' education, age group, and employment status. A random intercept was specified for each community cluster (j) to capture unobserved heterogeneity within the context. The model includes fixed effects for digital inclusion ($BDLIFT_{ij}$), residence ($Residence_{ij}$), their interaction ($BDLIFT_{ij} \times Residence_{ij}$), and individual-level controls: household wealth quintile, education level, age group, and current employment status. It employs a two-level mixed-effects logistic regression to predict the likelihood that a woman will report autonomy in her healthcare decisions. The β_0 is the intercept, and β_1 through β_7 are fixed-effect coefficients for the predictors. Sampling weights (wt) are applied at the individual level to adjust for the complex survey design of the 2022 BDHS, ensuring nationally representative estimates.

4 RESULTS AND ANALYSIS

This study uses the 2022 Bangladesh Demographic and Health Survey (BDHS-VIII) individual recode (IR) file (BDIR81FL), implemented by the National Institute of Population Research and Training (NIPORT) between June and December 2022 (BDIR81FL). The survey was nationally representative and applied a two-stage stratified cluster sampling design. A total of 30,078 ever-married women aged 15–49 were interviewed. For this analysis, the current study restricted the sample to women with complete information on the outcome variable (healthcare decision-making autonomy), the Bangladesh Digital Life Inclusion for Females and Technology (BD-LIFT) index, and all covariates. After excluding incomplete cases, the final analytic sample includes 15,112 women, weighted to ensure national representativeness. Table 2 presents the results from three multilevel logistic regression models examining women's autonomy in healthcare decision-making.

Table 2 presents the results from three multilevel logistic regression models examining women's autonomy in healthcare decision-making. Model 1 includes only the BD-LIFT index, showing the bivariate association between digital inclusion and independence. Model 2 adds an interaction between BD-LIFT and rural residence to test whether the effects of digital use vary by context. Model 3 incorporates the full set of socioeconomic and demographic controls, providing the most comprehensive specification. Across all models, the odds of reporting autonomy increase with higher levels of digital inclusion, and the full model confirms that these associations hold even after accounting for wealth, education, age, and employment status. Table 2 also indicates that women with full digital inclusion, such as device ownership, internet usage, and mobile money, have more than twice the odds of healthcare autonomy compared to those with no access. Small improvements are seen at intermediate BD-LIFT levels, while interactions with rural residence are weak and not statistically significant. Among controls, higher education, older age, and employment consistently predict autonomy, but wealth contributes little. Model fit improves in the full specification ($AIC = 15,815$; $BIC = 15,982$), and the intraclass correlation ($ICC = 0.083$) confirms modest but meaningful community-level variation. Overall, these findings highlight digital inclusion as a strong and independent predictor of women's healthcare decision-making, beyond traditional socioeconomic factors.

Table 1: Summary Statistics for the BD-LIFT Analytical models

Variable (role)	Current variable	Recoded from (DHS var)	Categories (n, %)	N
Dependent variable	autonomy_say (women's autonomy)	v743a	No say = 3,611 (23.9) Has a say = 11,501 (76.1)	15,112
Key independent variable	bdlift (digital inclusion index)	Constructed (v169a, v169c, v169b, v171a)	No inclusion = 9,521 (63.0) Device only = 1,007 (6.7) Device + Internet = 2,345 (15.5) Device + Internet + Mobile Money = 2,239 (14.8)	15,112
Moderator	Residence (urban–rural context)	v025	Urban = 5,270 (34.9) Rural = 9,842 (65.1)	15,112
Controls				
Wealth quintile	wealth (socioeconomic status)	v190	Poorest = 2,698 (17.9) Poorer = 2,911 (19.3) Middle = 2,931 (19.4) Richer = 3,079 (20.4) Richest = 3,493 (23.1)	15,112
Education	education (human capital)	v149	No education = 2,054 (13.6) Incomplete primary = 1,912 (12.7) Complete primary = 2,012 (13.3) Incomplete secondary = 5,068 (33.5) Complete secondary = 1,668 (11.0) Higher = 2,398 (15.9)	15,112
Age (10-year clusters)	age (life stage)	Clustered in 10 years (v012)	15–19 = 1,194 (7.9) 20–29 = 5,305 (35.1) 30–39 = 5,094 (33.7) 40–49 = 3,519 (23.3)	15,112
Employment status	is_working (economic empowerment)	v714	Not working = 10,622 (70.3) Working = 4,490 (29.7)	15,112

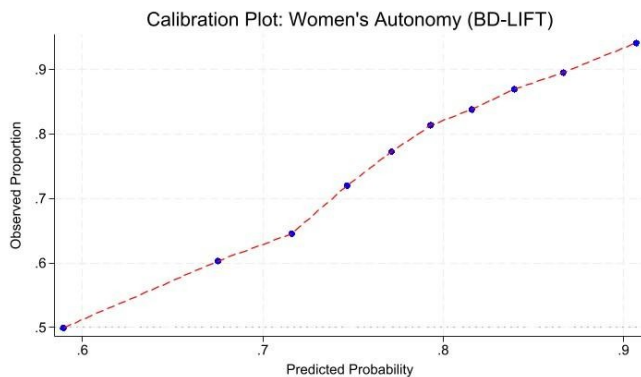
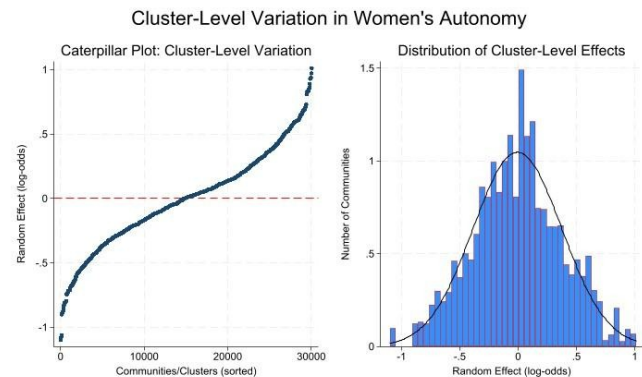
**Figure 2: Calibration Plot for the Full Model (Model 3)**

Figure 2 shows the calibration plot for the full model, demonstrating strong agreement between predicted probabilities and observed autonomy. Across ten prediction bins (0.6–0.9), observed proportions closely follow the 45° reference line, indicating accurate probability estimates. For example, a predicted probability of 0.7 corresponds to an observed rate of 0.68, and a predicted probability of 0.9 matches an observed rate of 0.92. Minor deviations

**Figure 3: Cluster level variation and impact (BD-Lift Model 3)**

happen in the mid-range, but overall, the pattern confirms that the BD-LIFT–based model reliably captures women's healthcare autonomy.

Figure 3 shows community-level differences in autonomy. The caterpillar plot displays clusters ranging from –1 to +1 log-odds, while the histogram resembles a normal distribution centered near

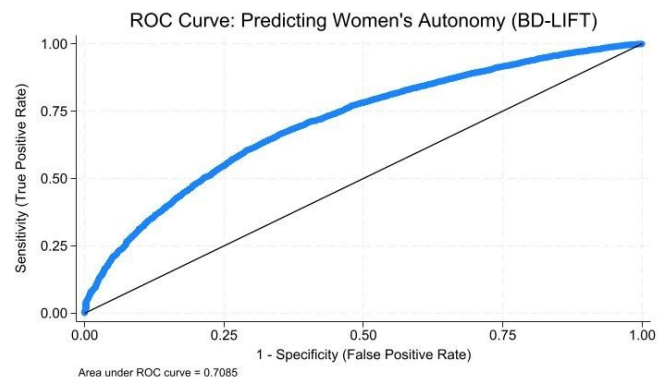
Table 2: Multilevel logistic regression of women's healthcare decision-making autonomy (Bangladesh DHS 2022).

Predictor	Model 1: Bivariate	Model 2: BD-LIFT × Residence	Model 3: Full Model
BD-LIFT (ref: no inclusion)			
Device only	1.20* (0.11)	1.23 (0.19)	1.30 (0.20)
Device + Internet	1.15* (0.07)	1.17 (0.12)	1.22 (0.13)
Device + Internet + Mobile money	2.04*** (0.17)	2.42*** (0.36)	2.35*** (0.37)
Residence (ref: urban)	—	0.92 (0.07)	0.92 (0.08)
Interaction (BD-LIFT × Rural)	—		
Device only × Rural	—	0.96 (0.18)	0.98 (0.19)
Device + Internet × Rural	—	0.96 (0.13)	1.01 (0.14)
Full inclusion × Rural	—	0.76 (0.14)	0.80 (0.15)
Wealth quintile (ref: poorest)	—	—	
Poorer	—	—	1.02 (0.08)
Middle	—	—	1.02 (0.08)
Richer	—	—	1.03 (0.09)
Richest	—	—	1.00 (0.10)
Education (ref: no education)	—	—	
Incomplete primary	—	—	1.10 (0.10)
Complete primary	—	—	1.13 (0.11)
Incomplete secondary	—	—	1.23* (0.10)
Complete secondary	—	—	1.28* (0.13)
Higher	—	—	1.30* (0.15)
Age group (ref: 15–19 years)	—	—	
20–29	—	—	1.44*** (0.12)
30–39	—	—	1.99*** (0.16)
40–49	—	—	2.10*** (0.20)
Employment (ref: not working)	—	—	
Working	—	—	1.35*** (0.07)
Constant	3.13*** (0.11)	3.32*** (0.22)	1.49** (0.19)
Random intercept variance (cluster)	0.30 (0.03)	0.29 (0.03)	0.30 (0.03)
Model fit			
Observations	15,112	15,112	15,112
Log likelihood	–7978.12	–7974.41	–7885.29
AIC	15,966	15,967	15,815
BIC	16,004	16,035	15,982

Notes: Odds ratios reported; robust standard errors in parentheses. Reference categories shown for clarity. * $p < .05$, ** $p < .01$, *** $p < .001$. ns = nonsignificant predictors retained for transparency. BD-LIFT = Bangladesh Digital Life Inclusion for Females & Technology index.

zero. The ICC (0.083) suggests that 8% of the variance occurs between communities, highlighting the role of local contexts alongside individual factors. Figure 5 presents the ROC curve (AUC = 0.71), indicating the model correctly identifies women with autonomy from those without in 71% of cases, demonstrating moderate predictive accuracy.

In model 3 the ROC curve (see Figure 4), classifies autonomy significantly better than chance (50%), confirming that BD-LIFT and other predictors enhance discriminating power. Figures 3–5 complement Table 2 and Figure 2 the calibration plot shows predicted and observed probabilities closely align, cluster-level plots reveal 8% of variance occurs between communities, and the ROC curve (AUC $\approx$ 0.71) indicates moderate discrimination. Together, these diagnostics confirm that BD-LIFT is both statistically robust and substantively meaningful in predicting women's healthcare autonomy in Bangladesh.

**Figure 4: ROC Curve for Full BD-Lift Model**

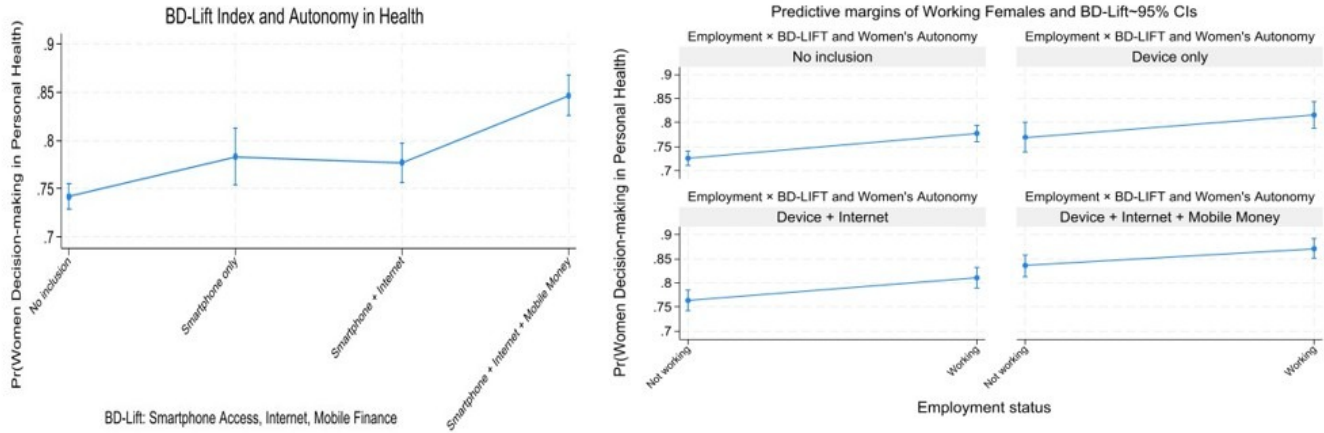


Figure 5: Predictive Margins of BD-Lift Index and employment on healthcare decision-making

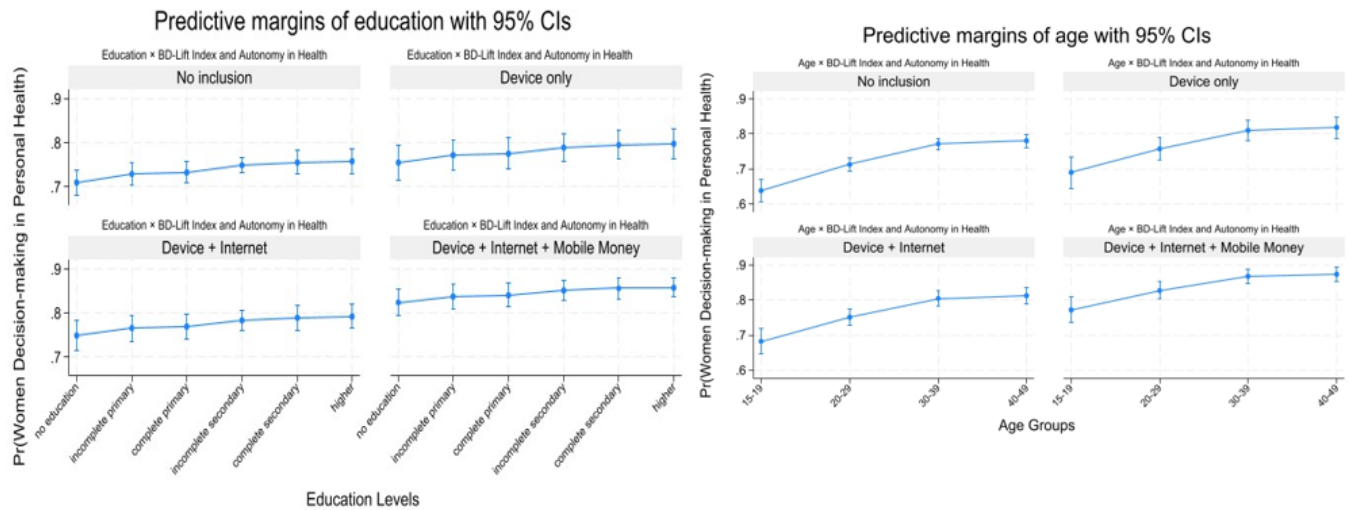


Figure 6: Predictive Margins of BD-Lift Index on Age and Educational Attainment

5 UNIQUE CONTRIBUTIONS

This current study examines the extent to which digital inclusion predicts women's autonomy in healthcare decisions in Bangladesh. The regression results in Table 2 and the coefficient plot (Figure 2) show that the BD-LIFT index is a consistent and strong predictor of autonomy, even after adjusting for education, wealth, employment, age, and place of residence.

Women with full digital inclusion—owning a phone, using the internet, and using mobile money—have more than twice the odds of reporting healthcare autonomy compared with women with no access. Education, older age, and employment are also positively associated, while wealth shows little effect. Predictive margins (Figure 5) show that average autonomy rises from about 74% among

women with no digital access to roughly 78% for those with a device-only or device-plus-internet connection, and to around 85–86% for women with full BD-LIFT inclusion. At each BD-LIFT level, working women have about 4–5 percentage points higher predicted autonomy than nonworking women, and working women with full digital inclusion reach nearly 87% predicted autonomy. Education adds a significant uplift (Figure 6): predicted autonomy rises from 74% among women with no schooling to 76% with incomplete primary, 77% with complete primary, and nearly 79% for those who finish secondary school or higher. This 5-point increase is smaller than the BD-LIFT shift but still meaningful, and it shows that education raises the baseline on which digital tools operate. These patterns match the regression odds ratios, which show significant effects for secondary and higher education.

Age adds a life-course perspective: autonomy increases from 67 percent among women aged 15–19 to 74 percent at 20–29 and about 80–81 percent in the 30s and 40s, a 13–14 point improvement that shows how maturity boosts agency while emphasizing the importance of digital inclusion to bridge early gaps. Working women have 35 percent higher odds of autonomy, highlighting the interaction between economic participation and life stage and digital access (see Figure 6). BD LIFT captures this trend more clearly than wealth or education, revealing a steep gradient in healthcare decision-making autonomy. For policy, the evidence suggests that ICTD and gender programs should expand connectivity with mobile finance and digital literacy to improve women's health agency, directly supporting both SDG 3 (health) and SDG 5 (gender equality).

6 LIMITATIONS

The current study has several limitations that should be acknowledged. First, the analysis relies on cross-sectional data from the 2022 Bangladesh Demographic and Health Survey (BDHS-VIII). The design allows us to estimate associations between digital inclusion and women's autonomy in healthcare decision-making, but it does not permit strong causal claims [27], [28], [29]. Causal claims remain limited because the data are observational: women with greater autonomy may also adopt digital tools more often, and this two-way link cannot be separated in cross-sectional BDHS 2022 data. Survey responses may also overstate autonomy in households where husbands or elders still make most decisions, especially in settings with strong paternalistic norms. The analytic sample includes 15,112 women with complete information on autonomy, BD-LIFT, and covariates, down from 30,078 in the full file; missingness is modest (<5%) but may still underrepresent women who are harder to reach or less connected. Model diagnostics indicate adequate but not strong predictive performance ($AUC = 0.71$), and the intraclass correlation ($ICC = 0.083$) suggests that most variation in autonomy is within rather than between communities. BD-LIFT covers three core parts of digital inclusion—smartphone ownership, internet use, and mobile money adoption—but not network quality, digital skills, social norms, or men's and household-level use, which may also shape women's power. Even with these limits, the analysis links a new cumulative index of digital inclusion to women's healthcare autonomy in a nationally representative sample: full BD-LIFT inclusion raises predicted autonomy from 74% to 86%, a 12-point gain that is larger than typical wealth or education effects, and the multilevel, well-documented workflow offers a clear, reproducible template for policy work on digital access and gendered health empowerment.

7 FUTURE RESEARCH

The current study empirically examines a pertinent research question: to what extent does digital inclusion predict women's autonomy in healthcare decision-making in Bangladesh? The study uses the 2022 Bangladesh Demographic and Health Survey, and a novel cumulative index titled BD-LIFT (Bangladesh Digital Life Inclusion for Females & Technology). It shows that digital inclusion has a strong and independent association with autonomy. Women with full digital access (device ownership, internet use, and mobile

money adoption) are more than twice as likely to have a say in their own healthcare compared to women with no digital access. The effect remains robust even after adjusting for wealth, education, employment, and residence. Through introducing the BD-LIFT index, this research extends prior work on autonomy in personal health decision making among women in Bangladesh and South Asia that has traditionally emphasized education, household resources, or cultural norms [15], [16], [17], [30], [31], [32], [33], [34], [35], [36]. Earlier studies have consistently shown that wealth and schooling are important determinants of women's decision-making [37], [38], [39], [40], yet the current findings demonstrate that digital access provides additional explanatory power that these traditional measures cannot capture. For example, household wealth quintiles show little independent effect in the full model, while BD-LIFT continues to predict autonomy with large, significant gains. This contrast underscores a key insight: in a digitalizing society, empowerment is increasingly tied not only to material resources but also to the ability to connect, transact, and communicate. The findings extend the DHS Framework for Analyzing Female Autonomy (Chapter 4), which draws on modernization theory to emphasize education, wealth, residence, and employment as core determinants of women's decision-making power [41], [41], [42], [43]. Prior DHS analyses show that women in urban areas, with higher education or employment, have greater autonomy, while those in rural or poorer households remain constrained. The current study confirms these patterns but also adds a new dimension: digital inclusion.

The results show that using the BD-LIFT index, Bangladeshi women with full digital inclusion, such as owning a device, using the Internet, and adopting mobile money, are more than twice as likely to report autonomy in healthcare ($OR = 2.35$) compared to women with no access. Even at intermediate levels of inclusion (device-only or device plus internet), autonomy increases by 15–30%. In contrast, community-level variation explains only 8.3% of the differences in autonomy ($ICC = 0.083$), indicating that digital access transcends local barriers and provides women with tools that traditional structural factors cannot fully capture. This both supports and challenges the DHS modernization framework, which largely views autonomy as a product of static socioeconomic variables. Digital tools change this perspective: they give women direct access to health information, resources, and decision-making power, even in settings where education or wealth are limited. For policy, the message is clear. Expanding smartphone access, affordable Internet, and mobile finance is not only an infrastructural issue but also a health and gender equality strategy. Investing in digital access can directly increase the number of women involved in healthcare decisions, helping to achieve Sustainable Development Goals 3 (health) and 5 (gender equality). Future research might extend the BD-LIFT index to other South Asian and African contexts, examine causal effects through longitudinal or intervention studies, and combine survey data with qualitative insights into how digital tools influence household negotiations. Overall, the evidence strongly supports the idea that digital inclusion is now essential to women's empowerment in Bangladesh, not just an add-on.

8 SUPPLEMENTARY MATERIAL

Supplementary material for this article includes extended regression tables, additional descriptive statistics, calibration and ROC diagnostics, and detailed predictive margins for BD-LIFT, education, age, and working status. These materials are provided as part of the replication package to support transparency and reuse.

9 DATA AVAILABILITY STATEMENT

Research documentation and data that support the findings of this study are openly available in the Harvard Dataverse at: Afzal, Muhammad Hassan Bin. 2025. "Replication Package for: From Connectivity to Care: How Digital Inclusion Shapes Women's Healthcare Decision-Making in Bangladesh." Harvard Dataverse. <https://doi.org/10.7910/DVN/DSQJU9>.

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